EXPERIMENT 8: USER MANAGEMENT AND ROLE-BASED ACCESS CONTROL IN ZOHO CREATOR

AIM

To implement User Management and Role-Based Access Control in Zoho Creator to manage users, assign roles, and control access to the features and data of the Student Attendance and Academic Performance Tracker based on the permissions granted to each role.

PROCEDURE

1. Log in to Zoho Creator using the administrator account.
2. Open the Users page from the Manage section in the navigation pane.
3. Click Add User and enter the email addresses of the users to be invited.
4. Verify that each user appears in the Users list with the correct name, email address, and status (Active or Pending).
5. Open the options menu beside a user to view the available actions, such as resending invitations, editing details, deactivating, or deleting a user.
6. Add users to the Student Attendance and Academic Performance Tracker application.
7. Assign the appropriate application role and permissions to each user.
8. Open the user's details and verify the application, role, and permission assigned.
9. Test each role to ensure that users can access only the permitted forms, reports, and features.
10. Deactivate a test user and verify that access is restricted.

ROLES PLANNED FOR THE APPLICATION

Role	User	Access Permissions
Administrator	College administrator	Full access to forms, reports, dashboards, users, and application settings
Faculty	Teaching staff	Add and edit attendance records and academic marks; view student reports
Attendance Staff	Attendance coordinator	Add and update attendance records; view attendance reports

Student	Individual student	View permitted academic and attendance information only
Department Head	Head of department	View department reports, performance charts, and low-attendance reports

PLANNED USERS (SAMPLE)

User	Sample Email	Role	Permission
Administrator	admin@example.com	Administrator	Full access
Faculty	faculty@example.com	Faculty	Attendance and marks management
Attendance Staff	attendance@example.com	Attendance Staff	Attendance entry and reports
Student	student@example.com	Student	View permitted personal records
Department Head	hod@example.com	Department Head	View department reports

Note: These email addresses are examples only. Replace them with actual user email addresses when configuring the application.

PROCESS DIAGRAM

Figure 8.1: Role-Based Access Control Flow

Administrator
↓
Add Users
↓
Assign Application Roles
↓
Configure Permissions
↓
Users Log In
↓
Access Forms and Reports According to Their Roles
↓
Verify Access Restrictions

ROLE TESTS

Role	Action	Expected Result

Administrator	Open all forms and reports	Allowed
Faculty	Add attendance and academic marks	Allowed
Attendance Staff	Update attendance records	Allowed
Student	View permitted personal records	Allowed
Student	Edit another student's marks	Not allowed
Department Head	Open department performance reports	Allowed
Deactivated User	Log in to the application	Access denied

RESULT

User Management and Role-Based Access Control were configured for the Student Attendance and Academic Performance Tracker in Zoho Creator. Users were assigned roles with appropriate permissions, and access restrictions were tested to ensure that application data and features could be accessed according to the assigned roles.

CONCLUSION

Role-Based Access Control in Zoho Creator helps protect student information by restricting access according to user responsibilities. The administrator can manage users, assign application roles, and control access to forms, reports, and dashboards. This improves data security, privacy, and the reliability of the Student Attendance and Academic Performance Tracker.